

Isaac Hoyt

I'm thrilled by the open-ended challenge to innovate particulate analysis within Alcon's product testing because I want to prove my worth as an engineer. I'm still trying to find what niche I'm most suited to as an engineer, and I would like to learn more about the manufacturing process in particular. I realize that developing new technology and prototypes is only one component of the impact that engineers can have. Manufacturing, however, is another critical step in bringing innovations to life and ensuring they meet real-world needs. By gaining a deeper understanding of this stage, I hope to identify where I can add the most value. I want to develop a product testing tool that allows for more efficient handpiece manufacturing, without sacrificing the quality of devices that leave the assembly line. For medical devices employed in ophthalmologic surgery, safety and reliability are of the utmost importance. I hope that the high standards required for this project push me to develop my capabilities as an engineer.

In addition, particle evaluation technology presents a uniquely exciting opportunity to me because it combines several engineering disciplines that have caught my attention. Because I'm interested in the science behind interferometry and imaging technologies, I look forward to investigating various imaging techniques and their utility as it relates to our engineering challenge. Experiences in instrumentation and data analysis, such as the 60 series of biomedical engineering classes, have equipped me for this kind of project. Combining the various components of a particulate analysis workflow will allow me to take advantage of the skills that I have procured throughout my coursework in biomedical engineering.

Tandol Mathew Jerthsattayanukul

I am a 4th year Biomedical engineering major working on an Alcon project for my senior design to develop a particulate counting device to test on the Alcon Hand-piece. When I first got introduced to the medical device called the Sentry Hand-piece that is used in the procedure of

eye surgery, I was drawn in right on the spot to help develop a particulate counting device that facilitates the job of technicians in the testing process of the hand-piece. The possibility of creating a functional device that is related to improving patients' lifestyle from suffering health complications brings joy to me.

I don't have any personal connection with this project except for the fact that a lot of my family members have passed away from diseases and health complications. Some health conditions have no definitive cure and that is one of the reasons I chose Biomedical engineering major in the first place. My ultimate goal is to create innovative medical device that would help people suffer less from health conditions, and potentially open up new spectrums in the medical field to cure the incurable disease

My goal is to get as much industry experience as possible. Being able to contribute in building particulate counting device will improve my engineering skills, therefore becoming a better engineer. Working on a project in this field will expose me to challenges, and develop my critical thinking skills to come up with solutions, making me become more versatile and a better problem solver. Main skills that I aim to improve upon are my software and designing skills.

I have completed a BME biomedical signals and systems course where we worked on a group project to create a code involving a bandpass filter that filters out the noise or artifact from raw data of patients' heart signals, diagnosing their arrhythmias. For particulate counting device, specifically the one designed to test the particles in a device like Sentry Hand-piece, implementing bandpass filter in the software can be very significant. Reducing the noises and focusing on the noise made by the particles in the device would make the testing system more precise and efficient. Another significant coursework that would be beneficial to this project is Solidworks. Having skill sets to design the particulate counting device and refine the design to maximize the functionality of the device will translate to the success and the efficiency of the development process.

Matthew Luong

For the Alcon: Automated Particulate Counting for Handpiece Production, I want to pursue this project because it provides knowledge in the industry and manufacturing process of a particular medical device. In this project, we will be learning with multiple types of engineers on the manufacturing of Alcon's Sentry Handpieces and how their quality is tested for any particulate in the devices. Unlike other industry projects, they do not go through the production process. Being involved in this project will help me move toward my career goals because it will provide me with industry experience, which I have not had the chance to experience during my college career. Only this year, I found experience working under a lab, but I am eager to find a career working for a medical device company. That requires industry experience, which I do not currently possess. During my time working under my research lab, I worked on multiple projects delving into biophotonics. I've learned how to conduct proper research into parts and materials and served as a software developer for both projects I participated in using Labview and Arduino. Most importantly, I learned how to apply my knowledge from the coding classes I took into a real-world problem, which requires self-learning to troubleshoot and build code from scratch. In addition, I have attended a 40+ hour workshop at Applied Medical, which is another medical device industry. Even though the experience was short, I thoroughly enjoyed the industry environment and desire to further involve myself in that field. The Alcon project matters to me because I desire to extend my knowledge outside of coding to be a more well-rounded engineer. The project has a role for a hardware and manufacturing position, which is what I want to be involved in. This project is a chance to provide me with experiences and skills that I have not yet obtained, which would be helpful going forward in my professional career. The biomedical engineering field is a competitive environment with lots of eager and determined students and engineers. It is my goal currently to gain skills and experiences to find my own niche in this forever-evolving field.

Brian Ngo

My fascination with biomedical engineering, especially its entrepreneurial aspects, has been a driving force throughout my academic journey. From a young age, I had ideas for startups—most of which were more imaginative than practical. But when my interests shifted toward medical innovations, the potential to make a meaningful impact became real. Currently, I conduct research in ophthalmology, where I am involved in a study to improve the ergonomics of ophthalmologists' postures during eye surgery. The opportunity to work on a project for Alcon resonated with me immediately, as it aligned closely with my interests and goals in the field.

The precision that Alcon brings to its devices reflects my own commitment to quality. "Good enough" has never felt adequate to me; I focus on even the smallest details because I believe people can sense quality and deserve the best. Alcon's dedication to achieving excellence in medical devices mirrors my personal standards, and I'm excited to work on a project that demands such rigor.

This project represents a step toward my goal of exploring the entrepreneurial side of biomedical engineering. While I have a solid foundation in biomedical engineering through academic coursework and research, I haven't yet had the chance to apply these skills independently. I look forward to gaining experience in quality assurance and learning what it takes to produce impactful medical devices at scale. I am particularly drawn to the potential redesign aspect of the project, which I hope will allow me to contribute innovative ideas to improve the device's efficiency and quality.

In terms of background, I currently conduct research for the ophthalmology department at UCI, where my PI has given me hands-on exposure to surgical tools, broadening my understanding and appreciation of ophthalmology. I also took a design course, "BME 60C: Engineering Analysis/Design: Computer-Aided Design," where I led a team project that highlighted my ability to delegate tasks, maintain team accountability, and support decision-making while fostering trust and independence. These skills were further developed in

a team competition I participated in from July to October 2024. This experience solidified my leadership capabilities and gave me a taste of what it's like to see a project through from start to finish.

Landon Stein

The projects I am currently working with are “Alcon: Automated Particulate Counting for Handpiece Production” and “Global Unmet Needs: Improving Microscope Visibility during Eye Surgeries in Paraguay.” For the Alcon project, I am interested in being more involved in the medical device development process. We get the opportunity to be intimately involved in the manufacturing process of Alcon’s handpieces which in the past have gone on to have applications within ophthalmology and eye surgeries. As for the Microscope Visibility Project, I am looking forward to it being a more hands-on project that will allow me to be directly involved in a project with clear real-world benefits.

In the past, I have worked with arterial catheters for use in the peripheral arteries and have gotten to work with research and development as well as quality. I have not yet had the opportunity to work in an ophthalmological setting and am excited for these two projects because they are both in a field which I have not yet had experience in. Ophthalmology is a very diverse field of study and I will just barely get to scratch the surface into what this field has to offer.

For most of my life I have never needed glasses or corrective lenses in order to see. However, last year, my vision started becoming blurrier than I had been used to, even causing mild nausea – obtaining glasses was now in my best interests for preserving my vision. I have always seen my friends with glasses without giving it much thought but after experiencing vision loss for myself I have an ability to better understand and empathize with what it is like. Being able to learn about vision, lenses, optics, and ophthalmology in general, is an exciting venture for me.

For working with Alcon, one of personal and career goals is to be a part in designing and producing medical devices that can help people, relieve pain, and increase quality of life for patients. Having the opportunity to work with such a renowned company feels like being another step closer to what my professional and personal goals strive to be.

As for working with Dr. Schmidt in Paraguay, I wanted an opportunity to have a very hands-on engineering experience – one that has very clear benefits to patients and people. I want an opportunity to showcase what I have learned in my studies and experiences and to use that to help others.

I enjoy designing with Solidworks and getting my hands dirty with building things – skills which I have spent a lot of time and effort on that I want to utilize in helping people. I am excited to work with all of my colleagues and I cannot wait to see how each of these projects and we as people evolve throughout the year. To release our projects and efforts to the world is a great opportunity and I hope that many people will be able to benefit from our works.

Yolanda Gomez

Alcon is a renowned biomedical device company dedicated to developing essential products for eye care. Vision is a fundamental part of human experience, shaping how we engage with the world, and when it's compromised, it can impact every part of our daily lives. One of Alcon's most impactful devices, the handpiece, has become an essential tool in ophthalmic surgeries which is used for everything from cataract tissue removal to retinal reattachment. The opportunity to contribute to advancing such a vital tool is incredibly motivating for me, and I am excited to learn more about the innovative thinking involved in optimizing devices that help restore vision for thousands of patients.

My interest in this field began when I was a child. When I was nine, my dad lost sight in one eye, which deeply affected him and our family. Known for his humor, even in tough times, he fell into a period of depression. In an attempt to cheer him up, I promised him I will build him

a bionic eye. He laughed it off, thinking it impossible. But the idea stuck with me, and I began reading about advancements in eye technology, realizing that a bionic eye might not be purely science fiction. This experience sparked my passion for biomedical engineering and my drive to contribute to innovations in eye care.

I believe that working on Alcon's projects will enhance my ability to think critically about optimizing medical devices and give me a firsthand understanding of the unique challenges involved in designing instruments for eye surgery. I'm eager to learn the engineering processes that support surgeons and maximize their ability to restore sight, as well as to gain insight into the industry I hope to join in the future. If my team and I succeed in this endeavor, I am confident it will make a meaningful impact, helping surgeons improve patients' quality of life.

For this project, I am ready to bring my research experience and technical skills in MATLAB, Arduino, SolidWorks, Inventor, and 3D printing. Additionally, I am proficient in laser cutting, soldering, and woodworking, with a solid foundation from courses in the BME 110, 60, 130, and 150 series. I am also excited to collaborate with talented minds in the industry, knowing that my technical knowledge combined with the expertise of my teammates will make us prepared to achieve our goals. I look forward to contributing to this journey, learning, and growing through hands-on experience in such an impactful field.